Diffie-Hellman Key Exchange

→ The Diffie-Hellman algorithm allows two or more parties to create a shared encryption key while communicating over an isecure network.

→ Communicating parties ~~sha~~ use the shared secret key for to create a unique symmetric key for encrypting and decrypting message.

* **Working of Diffie-Hellman:**

→ let 2 end users → Alice and Bob
↓
They must agree on positive whole integers p and q

→ p is prime number
→ q is generator of p

→ After Alice and Bob decided on p and q in secret, they select positive whole number personal keys a and b.

→ Alice and bob compute public keys a^* and b^* based on their personal keys using algorithms below:

$$a^* = q^a \bmod p$$
$$b^* = q^b \bmod p$$

→ Using their individual personal keys, each user can generate a number X from these public keys.

→ Alice uses following formula to Compute X-
$X = (b^*) \bmod p.$

Bob uses the following to Compute X.
$X = (a^*) \bmod p.$

Steps / Eg: Selecting the Prime Number & Generator

(1) Bob & Alice have decided on a prime number, $p = 23$

Also, they are also agree on a generator number, $g = 5$.

(2) Selecting Private keys.

Alice selects a secret number, for example $a = 16$.

Bob selects ~~numb~~ a secret number, for example ~~$b=15$~~ $b = 15$

(3) Determine the public keys

Alice first computes $A = g^a \bmod p$, which gives $A = 5^6 \bmod 23$, or 8

Bob uses formula $B = g^b \bmod p$, to get $B = 5^{15} \bmod 23$, or 19

(4) Exchange of Public keys

Alice provides Bob her computed public key, $A = 8$

Bob sends Alice his calculated public key, $B = 19$.

(5) Compute Shared Key

✓ Alice uses Bob's private key to compute the shared key –

$K_A = B^a \bmod p$ –

$K_A = 19^6 \bmod 23, = 2$.

✓ Bob uses Alice's public key to calculate the shared key – $K_B = A^b \bmod p$

$K_B = 8^{15} \bmod 23 = 2$.

∴ Alice & Bob can now encrypt their messages using same shared key, $K = 2$.